

Finding Zeros of Quadratic Functions Algebraically

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A **zero** of a quadratic function is the value of x that makes the function equal to zero.

Factored Form

A quadratic function in **factored form** is written as

$$f(x) = a(x - r_1)(x - r_2).$$

The zeros are r_1 and r_2 .

Zero Product Property

Use the **Zero Product Property**:

If

$$ab = 0,$$

then either $a = 0$ or $b = 0$.

Standard Form

A quadratic function in **standard form** is written as

$$f(x) = ax^2 + bx + c.$$

To find the zeros, solve

$$ax^2 + bx + c = 0.$$

Finding Zeros by Factoring

To find the zeros of a quadratic function in standard form, first solve

$$ax^2 + bx + c = 0$$

by **factoring** whenever possible.

Zeros and x -Intercepts

The zeros correspond to the x -**intercepts** of the graph of the quadratic function.

$$\text{zeros} = x\text{-intercepts}$$

Number of Zeros

A quadratic function may have:

two distinct zeros

one repeated zero

no real zero

Examples

Example 1

Find the zeros of

$$f(x) = (x - 5)(x + 2).$$

Set each factor equal to zero:

$$x - 5 = 0 \Rightarrow x = 5$$

$$x + 2 = 0 \Rightarrow x = -2$$

Therefore, the zeros are 5 and -2 .

Example 2

Find the zeros of

$$f(x) = 3(x + 1)^2.$$

Set the factor equal to zero:

$$x + 1 = 0$$

$$x = -1$$

Therefore, the repeated zero is -1 .

Example 3

Find the zeros of

$$f(x) = x^2 - 9x + 20.$$

Factor:

$$(x - 5)(x - 4) = 0$$

Therefore,

$$x = 5 \quad \text{or} \quad x = 4.$$

The zeros are 5 and 4.

Practice Exercises

Practice 1

Find the zeros of

$$f(x) = (x - 5)(x + 2).$$

Practice 2

Find the zeros of

$$f(x) = 2(x + 4)(x - 1).$$

Practice 3

Find the zeros of

$$f(x) = -(x - 6)(x - 3).$$

Practice 4

Find the zeros of

$$f(x) = 3(x + 1)^2.$$

Practice 5

Find the zeros of

$$f(x) = 5(x - 7)^2.$$

Practice 6

Find the zeros of

$$f(x) = x^2 - 9x + 20.$$

Practice 7

Find the zeros of

$$f(x) = 2x^2 - 7x + 3.$$

Practice 8

Find the zeros of

$$f(x) = x^2 + 5x + 6.$$

Practice 9

Find the zeros of

$$f(x) = 3x^2 - 12x + 9.$$

Practice 10

Find the zeros of

$$f(x) = 4x^2 + 4x - 15.$$

Activities

Activity 1

Find the zeros of

$$f(x) = (x - 4)(x + 3).$$

Activity 2

Find the zeros of

$$f(x) = 5(x - 2)(x + 1).$$

Activity 3

Find the zeros of

$$f(x) = -2(x - 8)(x - 5).$$

Activity 4

Find the zeros of

$$f(x) = 4(x - 3)^2.$$

Activity 5

Find the zeros of

$$f(x) = 2(x + 6)^2.$$

Activity 6

Find the zeros of

$$f(x) = x^2 - 11x + 28.$$

Activity 7

Find the zeros of

$$f(x) = 2x^2 - 5x - 3.$$

Activity 8

Find the zeros of

$$f(x) = x^2 + 7x + 12.$$

Activity 9

Find the zeros of

$$f(x) = 2x^2 - 8x + 8.$$

Activity 10

Find the zeros of

$$f(x) = 3x^2 + 2x - 8.$$

Thank you!